

9. FOLLOW-UP

Regular follow-up visits are recommended every 3 - 4 months in the first two years, every 6 - 8 months from subsequent years 3 - 5 and annually thereafter. The interval of visits should be adapted to the risk of relapse and patients' needs. The recommended surveillance are.¹³⁶

- annual ipsilateral (after BCS) and/or a contralateral mammography (after mastectomy), with US and breast MRI when needed
- regular bone density evaluation for patients on AIs or undergoing ovarian function suppression
- encouragement towards adopting a healthy lifestyle, including diet modification and exercise

Approximately 5% of breast cancer patients will have a risk to develop a new ipsilateral or contralateral cancer and it persists over time. There is limited evidence on surveillance strategies in older breast cancer survivors. The American Cancer Society has recommended screening mammography should be continued beyond 75 years old as long as a woman is in good health and is expected to live 10 more years or longer.¹³⁷ Nevertheless, current evidence is insufficient to assess the balance of benefits and harms of screening mammography in women aged 75 years or older.¹³⁸

10. SUPPORTIVE TREATMENT

- This chapter is written mainly based on the previous edition of CPG Management of Breast Cancer (Second Edition) and not on new clinical questions. The CPG DG opines that for complete management of breast cancer, the supportive treatment is summarised and mentioned here. Some updates are done on Psychosocial Assessment and Intervention Section.

a. Psychosocial assessment and intervention

Most breast cancer patients experience at least some psychosocial distress during the course of their diagnosis and treatment. A local study revealed that up to 47.1% of women with breast cancer experience psychological distress, depression (25.3%), anxiety (18.8%)¹³⁹ and post-traumatic stress disorder (19.6%).¹⁴⁰

Other common psychosocial concerns reported by women with breast cancer include:¹⁴¹

- fear of recurrence
- physical symptoms e.g. fatigue, sleep disturbance or pain
- body image disruption
- sexual dysfunction
- treatment-related anxieties
- intrusive thoughts about illness/persistent anxiety
- marital/partner communication
- feelings of vulnerability
- existential concerns regarding mortality

Local studies showed that psychosocial issues impact patients' overall well-being. In the currently challenging global economic state, almost half of Malaysian families with cancer survivors experienced financial constraints.¹⁴² Other related issues include lack of constant psychosocial support and proclivity of undesirable consequences e.g. delaying treatment and/or not turning up for treatment at all.^{143; 144; 145; 146; 147; 148; 149; 150}

Women with breast cancer should be screened for emotional distress using validated self-assessment psychological tests by trained healthcare providers.⁴ Psychological distress for them should be assessed at key time points throughout the cancer trajectory. This should be done at diagnosis, during and after completion of treatment throughout the survivorship period. Recommended screening tools for this purpose include Patient Health Questionnaire (PHQ) and Emotion Thermometer (ET). While the Hospital Anxiety and Depression Scale was previously widely used, it is now copyrighted and requires payment to use.

The aim of a psychological intervention is to reduce emotional and mental burden as well as to equip patients with accurate information and psychological skills to manage their distress. The following psychosocial interventions should be offered by trained healthcare providers particularly for women with emotional vulnerability following a diagnosis of breast cancer:⁴

- cognitive behaviour therapy
- psychosocial support
- psycho-education programmes (e.g. printed materials, audio-visual materials, telephone support and counselling)

Others include Managing Cancer and Living Meaningfully,^{151; 152} Mindfulness Therapy,¹⁵³ and Acceptance and Commitment Therapy.¹⁵⁴

b. Breast care nurse

Breast care nurse (BCN) improves the continuity of care and provide important information, support and referral for a wide range of needs experienced by breast cancer patients. All patients should have access to a BCN for treatment of breast cancer. Improving BCN accessibility includes:¹⁵⁵

- clarification on the role focusing on psychosocial assessment and support of the patients
- appropriate training, accreditation and ongoing professional development
- inclusion in MDT
- availability in rural and remote areas through telephone and video conferencing calls
- funding for positions should be ongoing and sustainable

A breast care nurse (BCN) should be assigned to all patients to provide information on management and psychosocial support throughout the diagnosis, treatment and follow-up.⁴

c. Lifestyle modifications

Lifestyle changes can be recommended to patients as an adjunct to standard breast cancer treatment.¹⁵⁶

The Malaysian breast cancer cohort study showed that there was a high proportion of breast cancer survivors being obese and overweight.^{148;}

¹⁵⁷ Breast cancer survivors should have balance diet and maintain lean body mass. There is still no particular style of diet that has been found to be more beneficial for reducing the risk of breast cancer recurrence.¹⁵⁶

Healthy lifestyle is associated with a lower risk of recurrence and this should include:

- achieving and maintaining a healthy weight through regular physical activity and dietary modification; regular physical activity

(at least 150 minutes/week) has the most robust effect of all lifestyle factors on reducing breast cancer recurrence¹⁵⁶

- limiting alcohol intake to below five units per week (3 or 4 alcoholic drinks per week or 6 g of alcohol per day)²⁵
- smoking cessation²⁵

d. Palliative care

Palliative care aims to improve quality of life of patients and their families in facing the life-threatening illness by effective pain management and other distressing symptoms. It also incorporates psychosocial and spiritual care according to patient/family needs, values, beliefs and cultures.¹⁵⁸ Palliative care is the main focus of care when cure is not achievable.²⁶

Palliative care begins at diagnosis of advanced breast cancer. It should be delivered concurrently with disease-directed therapies and facilitate patient's autonomy on the further management based on the medical information.

- Palliative care should be initiated by the primary care team and then augmented by a palliative care physician.

Essential components of palliative care include the following:²⁶

- establishment of rapport and relationships with patients and family/caregivers
- management of symptoms, psychosocial/spiritual distress and functional status
- exploration of patients' understanding and education about illness and prognosis
- clarification of treatment goals
- assessment and support of coping needs
- assistance with medical decision making
- coordination with other care providers
- provision of referrals to other care providers as indicated

e. Patient Navigation Programme

Patient navigation programme (PNP) is a community-based service delivery intervention designed to promote access to timely diagnosis and treatment of cancer and other chronic diseases by eliminating barriers to care.¹⁵⁹, level III

Principles of PNP are:

1. Navigation is a patient-centric healthcare service delivery model.
2. The core function of navigation is the elimination of barriers to timely care across all levels of healthcare which is most effectively carried out through a one-on-one relationship between the navigator and the patient.

3. Patient navigation should be defined to distinguish the role and responsibilities of the navigator from that of all other providers. Navigators should be integrated into the healthcare team in order to achieve maximum benefit for the individual patient.
4. Delivery of navigation services should be cost-effective and commensurate with the training and skills necessary to navigate an individual through a particular phase of the care.
5. The determination of whom should navigate should be primarily decided by the level of skills required at a given phase of navigation ranging from trained lay investigators to professional providers.
6. There is a need to define the point where navigation begins and ends; the need is not over until the cancer is resolved.
7. There is a need to navigate patients across disconnected level of care i.e. from primary to tertiary care.
8. Navigation systems require coordination. In larger systems of patient care, the coordination is best carried out by assigning a navigation coordinator or champion who is responsible for overseeing all phases of navigation activity within a given healthcare level.

PNP has the potential to reduce cancer-related disparities and improve outcomes by eliminating barriers to obtain quality cancer care. PNP in cancer care is effective in improving screening rates, adherence to follow-up following an abnormal results and timeliness of diagnosis.^{159, level III} Minimal research has indicated that PNP is effective for post-treatment surveillance.^{160, level I}

Malaysia is one of the countries with lower relative survival rate in the Asia-Pacific, with only 66.8% of patients having 5-year survival rate, mainly due to late presentation and poor adherence to evidence-based treatment.^{2, level II-2}

Studies in psychosocial factors show that fear, poor health education and lack of empowerment among Malaysians are major reasons causing delay and defaulting treatment. In order to address these challenges, Cancer Research Malaysia identifies the PNP as a potential community-based solution to improve the Malaysian survivorship of breast cancer.¹⁶¹

f. Breast cancer patient support groups

Breast cancer patient support groups are invaluable resources for support, services and information which include:

- provision of emotional, social and material support for individuals with breast cancer
- empowering breast cancer patients to self-care during and after treatment
- public education on breast cancer awareness

Breast cancer patient support groups play a crucial role in making patients adapted with their disease. A case-control study showed that patients supported by a peer group enjoyed a higher quality of life compared with others.^{162, level II-2} Locally, the breast cancer support groups are such as:

- Breast Cancer Welfare Association Malaysia
- Pink Unity (National Cancer Society Malaysia)
- Breast Cancer Foundation (formerly known as Pride Foundation)
- Pink Ribbon
- KanWork

- The following additional management is important and should be considered for breast cancer patients when indicated:
 - psychosocial assessment and intervention
 - breast care nurse
 - lifestyle modifications
 - palliative care
 - patient navigation programme
 - breast cancer patient support groups

Refer to **Appendix 9 on Post-Treatment Cancer Survivorship (Management of Treatment Complications)**

11. IMPLEMENTING THE GUIDELINES

The management of breast cancer should be guided by evidence-based approach in order to provide quality care to the patients. Several factors may affect the implementation of recommendations in the CPG.

11.1 Facilitating and Limiting Factors

Existing facilitators for application of the recommendations in the CPG include:

- a. wide dissemination of the CPG to healthcare providers (hard- and soft-copies)
- b. regular topic update for healthcare providers via continuous medical education (seminar/conference/course)
- c. National Cancer Registry
- d. National Key Performance Indicator i.e. margin on BCS
- e. involvement of non-governmental organisations e.g. breast cancer support groups in Breast Cancer Awareness Month

Existing barriers for application are:

- a. lack of understanding/limited knowledge on breast cancer
- b. insufficient resources including expertise, diagnostic tools, medications, equipment
- c. variation in clinical management and preferences

11.2 Potential Resource Implications

To implement the CPG, there must be strong commitment to:

- a. ensure widespread distribution of CPG & its implementation strategies
- b. strengthen training to ensure up-to-date information being shared
- c. provide adequate resources in the management of breast cancer
- d. provide multidisciplinary team at different levels of care
- e. strengthen the cancer registry
- f. empower community with active involvement in disease-related activities

The following is proposed as clinical audit indicator for quality management of breast cancer:

$$\text{Percentage of patients with suspected breast cancer* referred within two weeks to the breast clinic**} = \frac{\text{Number of patients with suspected breast cancer referred within two weeks to the breast clinic in a period}}{\text{Number of patients with suspected breast cancer in the same period}} \times 100\%$$

*Women aged >35 years with signs and symptoms, high risk group with signs and symptoms and patients with clinical signs of malignancy

**Target of 80%

$$\text{Percentage of breast cancer patients with clear surgical margins in BCS***} = \frac{\text{Number of breast cancer patients with clear surgical margins in BCS in a period}}{\text{Number of breast cancer patients with BCS in a period}} \times 100\%$$

***Target of 85%

Implementation strategies will be developed following the approval of the CPG by MoH which include launching of the CPG, Quick Reference and Training Module.

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Appendix 1

EXAMPLE OF SEARCH STRATEGY

Clinical Question: What is the adequate tumour free margin in breast conserving surgery in breast cancer?

1. BREAST NEOPLASMS/
2. (breast adj1 (cancer or carcinoma* or neoplasm* or tumo*)).tw.
3. (breast malignant adj2 (neoplasm* or tumo*)).tw.
4. (human mammary adj2 (cancer or carcinoma* or neoplasm* or tumo*)).tw.
5. cancer of breast.tw.
6. cancer of the breast.tw.
7. malignant neoplasm of breast.tw.
8. malignant tumo?r of breast.tw.
9. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8
10. MASTECTOMY, SEGMENTAL/
11. breast quadrantectomy*.tw.
12. breast conserv* adj2 therap*.tw.
13. ((breast conserv* or breast-conserv* or breast sparing or breast-sparing) adj2 surgery*).tw.
14. lumpectomy*.tw.
15. ((partial or segmental) adj1 mastectomy*).tw.
16. segmentectomy*.tw.
17. wide local excision.tw.
18. 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17
19. 9 and 18
20. tum?r margin.tw.
21. 19 and 20
22. limit 21 to (english language and humans and yr="2010 -Current")

Appendix 2

CLINICAL QUESTIONS

A. Risk factors

- What are the risk factors for breast cancer?

B. Screening

- What are the effective screening methods of breast cancer in:
 - general population?
 - high risk group?

C. Referral

- What are the criteria for referral to surgical or breast clinic?

D. Assessment/Diagnosis

- What are the effective approaches for imaging-histopathology discordance in breast lesions?
- What is the accuracy of tomosynthesis in the diagnosis of breast cancer?
- What is the accuracy of imaging-guided biopsy in breast lesions?
- What are the elements of adequate imaging report for breast cancer?
- What is the accuracy of Ki-67 as a predictor and prognostic marker in breast cancer?
- What is the accuracy of HER2 test in core biopsy specimen of breast cancer?

E. Staging

- Is positron emission tomography (PET) or PET/computed tomography accurate and effective in the staging of breast cancer?

F. Treatment (based on stage)

- What is the role of multidisciplinary team in breast cancer?
- What is the adequate tumour free margin in breast conserving surgery in breast cancer?
- Is neoadjuvant systemic therapy effective and safe for breast conserving surgery in breast cancer?
- What is the effective and safe subsequent treatment after sentinel lymph node biopsy in early breast cancer?
- Is surgery of primary tumour effective and safe in metastatic breast cancer?
- When is the optimal timing for breast reconstruction (with or without prosthesis) in breast cancer requiring post-operative radiotherapy?
- What is the most effective and safe taxane-based regimen in early breast cancer?
- Is anti-HER2 treatment effective and safe in neoadjuvant, adjuvant and metastatic breast cancer?
- Is subcutaneous trastuzumab effective and safe in breast cancer?

- Is ovarian suppression/ovarian ablation effective and safe in premenopausal breast cancer?
- Is endocrine therapy effective and safe in breast cancer?
- Is intraoperative radiotherapy/partial breast irradiation effective and safe in breast cancer?
- What is the effective and safe first-line systemic therapy in metastatic breast cancer?
- How to monitor the risk of osteoporotic fractures in patients with breast cancer on aromatase inhibitors?
- Are bone-modifying agents effective and safe in adjuvant and metastatic breast cancer?
- Which group of breast cancer patients need to be referred for fertility preservation?
- What are the effective and safe interventions for fertility preservation in breast cancer?
- What is the effective and safe local treatment of metastatic breast cancer?
- What is the effective and safe treatment in loco-regional recurrent breast cancer?

G. Survivorship Programme

- When to stop surveillance mammography in breast cancer?

H. Familial Breast Cancer

- Who should be offered genetic counselling and testing for inherited risk to hereditary breast and ovarian cancer? When is the optimal timing to do it?
- What are the effective clinical management strategies for those with inherited risk of hereditary breast and ovarian cancer?
- What are the effective clinical management strategies in BRCA carriers?

Appendix 3

RECOMMENDED REPORTING SYSTEM

INTRODUCTION

The ACR BI-RADS Atlas 2013 (4) is the updated version of the 2003 Atlas. BI-RADS® is designed to standardise breast imaging reporting and to reduce confusion in breast imaging interpretations. It also facilitates outcome monitoring and quality assessment.

It contains a lexicon for standardised terminology (descriptors) for mammography, breast US and MRI, as well as chapters on Report Organisation and Guidance for use in daily practice.

A. REPORT ORGANISATION

A good reporting system should be concise and organised using the following structure.

THE STANDARD REPORTING SYSTEM:

A statement indicating that the current examination has been compared to previous examination(s) should be included (specify date[s]). If this is not included, it should be assumed that no comparison has been made, although it is preferable to indicate that no comparison was made.

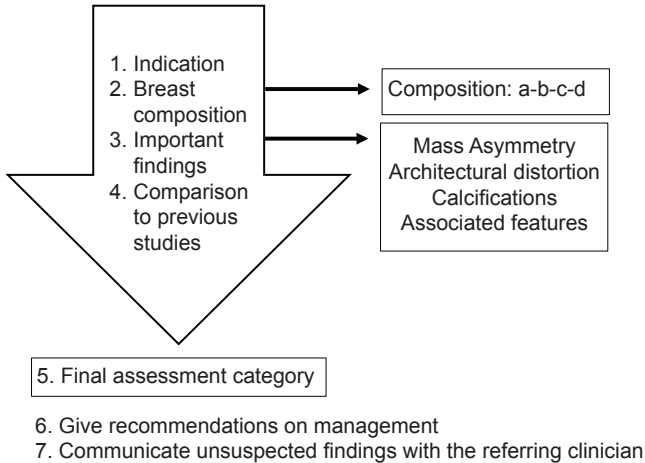
Below are the components of standard reporting system:

1. *Describe the indication for the study.*
2. *Describe the breast composition.*
3. *Describe any significant finding using standardised terminology.*
4. *Compare to previous studies.*
5. *Conclude to a final assessment category.*

Use BI-RADS categories 0 - 6 and the phrase associated with them. If mammography and US are performed: overall assessment should be based on the most abnormal of the two breasts, based on the highest likelihood of malignancy.

6. *Give management recommendations.*
7. *Communicate unexpected findings with the referring clinician.*

Standard Reporting



1. Describe the indication for the study

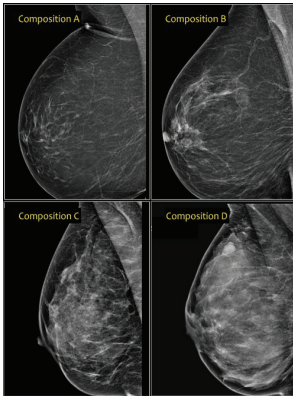
The examination is for screening, diagnostic or follow-up. Mention patient's history.

If US is performed, mention if it is targeted to a specific location or supplementary screening. If an implant is present, both standard and implant-displaced views should be performed and, this should be stated in the mammography report.

2. Describe the breast composition

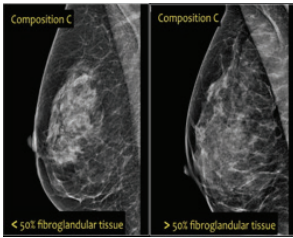
In the BI-RADS 2003 edition, the assignment of the breast composition was based on the overall density resulting in ACR category 1 (<25% fibroglandular tissue), category 2 (25 - 50%), category 3 (50 - 75%) and category 4 (>75%).

In BI-RADS 2013, the use of percentages is discouraged because in individual cases, it is more important to take into account the chance that a mass can be obscured by fibroglandular tissue than the percentage of breast density as an indicator for breast cancer risk.



In the BI-RADS edition 2013, the assignment of the breast composition is changed into a, b, c and d-categories followed by a description:

- a- *The breasts are almost entirely fatty.*
Mammography is highly sensitive in this setting.
- b- *There are scattered areas of fibroglandular density.*
The term density describes the degree of x-ray attenuation of breast tissue but not discrete mammographic findings.
- c- *The breasts are heterogeneously dense, which may obscure small masses.*
Some areas in the breasts are sufficiently dense to obscure small masses.
- d- *The breasts are extremely dense, which lowers the sensitivity of mammography.*



Notice in the left example, the composition is c - heterogeneously dense, although the volume of fibroglandular tissue is less than 50%.

The fibroglandular tissue in the upper part is sufficiently dense to obscure small masses. So, it is called c, because small masses can be obscured.

Historically this would have been called an ACR 2: 25 - 50% density.

The example on the right has more than 50% glandular tissue and is also called composition c.

3. Describe any significant finding using standardized terminology

Use the morphological descriptors: mass, asymmetry, architectural distortion and calcifications.

These findings may have associated features, like for instance a mass can be accompanied with skin thickening, nipple retraction, calcifications, etc.

Correlate these findings with the clinical information, mammography, US or MRI.

Integrate mammography and US findings in a single report.

a. Mass:

Size

Morphology (shape, margin)

Density

Associated calcifications

Associated features

Location

b. Calcifications:

Morphology - describe typically benign or malignant type by describing the shape of particles

Distribution

Associated features

c. Architectural Distortion:

Associated calcifications

Associated features

Location

d. Asymmetries (asymmetry, global asymmetry, focal asymmetry, developing asymmetry):

Associated calcifications

Associated features

Location

e. Intramammary lymph node (rarely important):

Location

f. Skin lesion (rarely important):

Location

g. Solitary dilated duct (rarely present):

Location

Mammography and US Lexicon

The table below shows a summary of the mammography and US lexicon.

Enlarge the table by clicking on the image.

First describe the breast composition.

When there is a significant finding, use the descriptors in the table.

The US lexicon has many similarities to the mammography lexicon, but there are some descriptors that are specific for US.

Mammography Lexicon				Ultrasound Lexicon			
Breast composition	A. entirely fatty B. scattered areas of fibroglandular density C. heterogeneously dense, which may obscure masses D. extremely dense, which lowers sensitivity			Breast composition	a. homogenous - fat b. homogenous - fibroglandular c. heterogenous		
Mass	shape	oval - round - irregular		Mass	shape	oval - round - irregular	
	margin	circumscribed - onsured - microlobulated - indistinct - spiculated			margin	circumscribed or not circumscribed: indistinct, angular, microlobulated, spiculated	
	density	fat - low - equal - high			orientation	parallel - not parallel	
Asymmetry	asymmetry - global - focal - developing				echo pattern	anechoic - hyperechoic - complex cystic/solid hypoechoic - isoechoic - heterogenous	
Architectural distortion	distorted parenchyma with no visible mass				posterior features	no features - enhancement - shadowing - combined pattern	
Calcifications	morphology	typically benign		Calcifications	in mass - outside mass - intraductal		
		suspicious	1. amorphous 2. coarse heterogenous 3. fine pleiomorphic 4. fine linear or fine linear branching	Associated features	architectural distortion - duct changes - skin thickening - skin retraction - edema - vascularity (absent, internal, rim) - elasticity		
	distribution		diffuse - regional - grouped - linear - segmental				
Associated features	skin retraction - nipple retraction - skin thickening - trabecular thickening - axillary adenopathy - architectural distortion - calcifications			Special cases (cases with a unique diagnosis)	simple cyst - clustered microcysts - complicated cyst - mass in or on skin - foreign body (including implants) - intramammary lymph node - AVM - Mondor disease - postsurgical fluid collection - fat necrosis		

US-Breast Imaging Lexicon

Many descriptors for US are the same as for mammography. For instance when describe the shape or margin of a mass.

Below are findings that are specific for US:

Breast composition:

- Homogeneous echotexture-fat
- Homogeneous echotexture-fibroglandular
- Heterogeneous echotexture

Mass:

- Orientation: unique to US-imaging, and defined as parallel (benign) or not parallel (suspicious finding) to the skin.
- Echo pattern: anechoic, hypoechoic, complex cystic and solid, isoechoic, hyperechoic, heterogeneous.
Echogenicity can contribute to the assessment of a lesion, together with other feature categories. Alone it has little specificity.
- Posterior features: enhancement, shadowing.
Posterior features represent the attenuation characteristics of a mass with respect to its acoustic transmission, also of additional value. Alone it has little specificity.

Calcifications:

- On US poorly characterised compared with mammography, but can be recognised as echogenic foci, particularly when in a mass.

Special cases - cases with a unique diagnosis or pathognomonic

US appearance:

- Simple cyst
- Complicated cyst
- Clustered microcysts
- Mass in or on skin
- Foreign body including implants
- Lymph nodes - intramammary
- Lymph nodes- axillary
- Vascular abnormalities
- Post-surgical fluid collection
- Fat necrosis

Correlate these findings with clinical information or findings of mammography, US or MRI. Integrate the mammography and US findings in a single report.

4. Comparison to previous examination(s), if deemed appropriate by the interpreting physician

Awaiting to compare with previous examination may assume importance if the finding of concern requires an evaluation of change or stability. Comparison is not important when a finding has unequivocally benign features. Comparison may be irrelevant when the finding is inherently suspicious for malignancy.

5. **Conclude to final assessment category**

The BI-RADS® assessment categories are designed to be concordant with specific management recommendations. The linking of assessment categories with concordant management recommendations further enhances sound medical practice.

If mammography and US are performed; the overall assessment should be based on the most abnormal of the two breasts, based on the highest likelihood of a malignancy.

An incomplete (category 0) assessment is usually given for screening examinations when additional imaging evaluation is recommended before it is appropriate to render a final assessment.

In category 0 assessment should include specific suggestions for the next course of action (spot-compression magnification views, US etc)

Final Assessment Categories			
Category		Management	Likelihood of Cancer
0	Need additional imaging or prior examinations	Recall for additional imaging and/or await prior examinations	Not applicable
1	Negative	Routine screening	Essentially 0%
2	Benign	Routine screening	Essentially 0%
3	Probably benign	Short interval follow-up (six months) or continued surveillance	>0% but ≤2%
4	Suspicious	Tissue diagnosis	4a. low suspicion of malignancy (>2% to ≤10%) 4b. moderate suspicion of malignancy (>10% to ≤50%) 4c. high suspicion of malignancy (>50% to <95%)
5	Highly suggestive of malignancy	Tissue diagnosis	≥95%
6	Known biopsy-proven	Surgical excision when clinical appropriate	Not applicable

6. **Give management recommendations.**

7. **Communicate unexpected findings with referring clinician.**

The verbal discussions between radiologist, patient and referring clinician should be documented in the clinical notes/report.

Adapted: Radiology Assistant (Available at: <https://radiologyassistant.nl/breast/bi-rads-for-mammography-and-ultrasound-2013>)

Appendix 4

BREAST IMAGING SURVEY FORM

JABATAN RADIOLOGI

Hospital _____

Tel: _____ Ext: _____ / _____ or NGCS _____

BREAST IMAGING SURVEY FORM (Please fill in into two copies)

RN: _____ IC/Passport No: _____

Name: _____

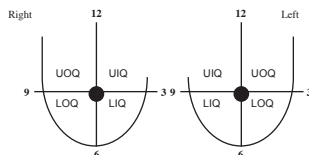
Request: ☐ Screening ☐ Diagnostic ☐ Additional Views

Previous imaging & date: _____

Menarche ☐ y/o Menarche ☐ y/o LMP / /
Day Month YearParity ☐ Number of children ☐ Breastfed (in months)

Risk Factors					Remarks
Family History of Cancer (Relationship & age of onset)					
Personal History of Cancer (Breast, Ovarian and others)					
Hormonal history (HRT/OCP and others)					
Genetic testing (BRCA 1, BRCA 2 and others)					
Clinical Data	Right		Left		Remarks
	Yes	No	Yes	No	
Breast pain/tenderness					
Lump in breast					
Nipple discharge					
Skin & nipple changes					
Nipple retraction/inversion					
Axillary nodes swelling					
Biopsy history & HPE					
Previous surgical intervention (Surgery/Implant/RT/ChemoTx)					

Impression:

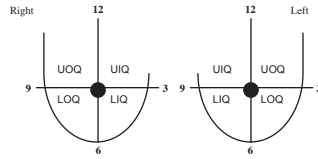


Signature & stamp of the Medical Officer/Specialist

RADIOGRAPHER FINDINGS:-

Please note any:

Scar	
Mole	
Lump	
Nipple changes	
Skin Folds	



Projection	kVp	mAs	Thickness	Dose	No. of Images	PGMI score
Right						
CC						
MILO						
Left						
CC						
MILO						
Additional views						
Repeat projections						

BIRADS Breast composition:

a	b	c	d	e
---	---	---	---	---

Signature of Radiographer)**RADIOLOGIST REPORT:-**

Please tick (✓) where appropriate

Findings	RT	LT
Mass		
Architectural distortion		
Asymmetrical density		
Calcification :		
(i) Macro		
(ii) Micro		
Axillary nodes		
Others		

Impression:

BIRADS Category:-

0	1	2	3	4	5	6
---	---	---	---	---	---	---

Recommendation:-

Signature & stamp of the Medical Officer/Specialist

Source: Ministry of Health Malaysia. Makluman Penggunaan Borang Permohonan Pemeriksaan Radiologi bagi Perkhidmatan Radiologi Yang Baharu Di Fasiliti Kementerian Kesihatan Malaysia. Putrajaya 6 November 2018

Appendix 5

TNM CLASSIFICATION OF BREAST CANCER

Definition of Primary Tumor (T) – Clinical and Pathological

T Category	T Criteria
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis (DCIS)*	Ductal carcinoma in situ
Tis (Paget)	Paget disease of the nipple NOT associated with invasive carcinoma and/or carcinoma in situ (DCIS) in the underlying breast parenchyma. Carcinomas in the breast parenchyma associated with Paget disease are categorized based on the size and characteristics of the parenchymal disease, although the presence of Paget disease should still be noted.
T1	Tumor ≤ 20 mm in greatest dimension
T1mi	Tumor ≤ 1 mm in greatest dimension
T1a	Tumor > 1 mm but ≤ 5 mm in greatest dimension (round any measurement 1.0-1.9 mm to 2 mm).
T1b	Tumor > 5 mm but ≤ 10 mm in greatest dimension
T1c	Tumor > 10 mm but ≤ 20 mm in greatest dimension
T2	Tumor > 20 mm but ≤ 50 mm in greatest dimension
T3	Tumor > 50 mm in greatest dimension
T4	Tumor of any size with direct extension to the chest wall and/or to the skin (ulceration or macroscopic nodules); invasion of the dermis alone does not qualify as T4
T4a	Extension to the chest wall; invasion or adherence to pectoralis muscle in the absence of invasion of chest wall structures does not qualify as T4
T4b	Ulceration and/or ipsilateral macroscopic satellite nodules and/or edema (including peau d'orange) of the skin that does not meet the criteria for inflammatory carcinoma
T4c	Both T4a and T4b are present
T4d	Inflammatory carcinoma (see "Rules for Classification")

*Note: Lobular carcinoma in situ (LCIS) is a benign entity and is removed from TNM staging in the AJCC Cancer Staging Manual, 8th Edition.

Definition of Regional Lymph Nodes - Clinical (cN)

cN Category	cN Criteria
cNX*	Regional lymph nodes cannot be assessed (e.g., previously removed)
cN0	No regional lymph node metastases (by imaging or clinical examination)
cN1	Metastases to movable ipsilateral Level I, II axillary lymph node(s)
cN1mi"	Micrometastases (approximately 200 cells, larger than 0.2 mm, but none larger than 2.0 mm)
cN2	Metastases in ipsilateral Level I, II axillary lymph nodes that are clinically fixed or matted; or in ipsilateral internal mammary nodes in the absence of axillary lymph node metastases
cN2a	Metastases in ipsilateral Level I, II axillary lymph nodes fixed to one another (matted) or to other structures
cN2b	Metastases only in ipsilateral internal mammary nodes in the absence of axillary lymph node metastases
cN3	Metastases in ipsilateral infraclavicular (Level III axillary) lymph node(s) with or without Level I, II axillary lymph node involvement; or in ipsilateral internal mammary lymph node(s) with Level I, II axillary lymph node metastases; or metastases in ipsilateral supraclavicular lymph node(s) with or without axillary or internal mammary lymph node involvement
cN3a	Metastases in ipsilateral infraclavicular lymph node(s)
cN3b	Metastases in ipsilateral internal mammary lymph node(s) and axillary lymph node(s)
cN3c	Metastases in ipsilateral supraclavicular lymph node(s)

Note: (sn) and (f) suffixes should be added to the N category to denote confirmation of metastasis by sentinel node biopsy or fine needle aspiration/core needle biopsy respectively.

*The cNX category is used sparingly in cases where regional lymph nodes have previously been surgically removed or where there is no documentation of physical examination of the axilla.

"cN 1mi is rarely used but may be appropriate in cases where sentinel node biopsy is performed before tumor resection, most likely to occur in cases treated with neoadjuvant therapy.

Definition of Regional Lymph Nodes - Pathological (pN)

pN Category	pN Criteria
pNX	Regional lymph nodes cannot be assessed (e.g., not removed for pathological study or previously removed)
pN0	No regional lymph node metastasis identified or ITCs only
pN0(i+)	ITCs only (malignant cell clusters no larger than 0.2 mm) in regional lymph node(s)
pN0(mol+)	Positive molecular findings by reverse transcriptase polymerase chain reaction (RT-PCR); no ITCs detected
pN1	Micrometastases; or metastases in 1 - 3 axillary lymph nodes; and/or clinically negative internal mammary nodes with micrometastases or macrometastases by sentinel lymph node biopsy
pN1mi	Micrometastases (approximately 200 cells, larger than 0.2 mm, but none larger than 2.0 mm)
pN1a	Metastases in 1 - 3 axillary lymph nodes, at least one metastasis larger than 2.0 mm
pN1b	Metastases in ipsilateral internal mammary sentinel nodes, excluding ITCs
pN1c	pN1a and pN1b combined
pN2	Metastases in 4 - 9 axillary lymph nodes; or positive ipsilateral internal mammary lymph nodes by imaging in the absence of axillary lymph node metastases
pN2a	Metastases in 4 - 9 axillary lymph nodes (at least one tumor deposit larger than 2.0 mm)
pN2b	Metastases in clinically detected internal mammary lymph nodes with or without microscopic confirmation; with pathologically negative axillary nodes
pN3	Metastases in 10 or more axillary lymph nodes; or in infraclavicular (Level III axillary) lymph nodes; or positive ipsilateral internal mammary lymph nodes by imaging in the presence of one or more positive Level I, II axillary lymph nodes; or in more than three axillary lymph nodes and micrometastases or macrometastases by sentinel lymph node biopsy in clinically negative ipsilateral internal mammary lymph nodes; or in ipsilateral supraclavicular lymph nodes
pN3a	Metastases in 10 or more axillary lymph nodes (at least one tumor deposit larger than 2.0 mm); or metastases to the infraclavicular (Level III axillary lymph) nodes
pN3b	pN1a or pN2a in the presence of cN2b (positive internal mammary nodes by imaging); or pN2a in the presence of pN1b
pN3c	Metastases in ipsilateral supraclavicular lymph nodes

Note: (sn) and (f) suffixes should be added to the N category to denote confirmation of metastasis by sentinel node biopsy or FNA/core needle biopsy respectively, with NO further resection of nodes.

Definition of Distant Metastasis (M)

M Category	M Criteria
M0	No clinical or radiographic evidence of distant metastases*
cM0(i+)	No clinical or radiographic evidence of distant metastases in the presence of tumor cells or deposits no larger than 0.2 mm detected microscopically or by molecular techniques in circulating blood, bone marrow, or other nonregional nodal tissue in a patient without symptoms or signs of metastases
M1	Distant metastases detected by clinical and radiographic means (cM) and/or histologically proven metastases larger than 0.2 mm (pM)

*Note that imaging studies are not required to assign the cM0 category

AJCC Anatomic Stage Groups

The Anatomic Stage Group table should only be used in global regions where biomarker tests are not routinely available. Cancer registries in the U.S. must use the Prognostic Stage Group table for case reporting.

When T is...	And N is...	And M is...	Then the stage group is...
Tis	N0	M0	0
T1	N0	M0	IA
T0	N1mi	M0	IB
T1	N1mi	M0	IB
T0	N1	M0	IIA
T1	N1	M0	IIA
T2	N0	M0	IIA
T2	N1	M0	IIB
T3	N0	M0	IIB
T0	N2	M0	IIIA
T1	N2	M0	IIIA
T2	N2	M0	IIIA
T3	N1	M0	IIIA
T3	N2	M0	IIIA
T4	N0	M0	IIIB
T4	N1	M0	IIIB
T4	N2	M0	IIIB
Any T	N3	M0	IIIC
Any T	Any N	M1	IV

Source: American Joint Committee on Cancer. AJCC Cancer Staging Manual, Eighth Edition. Springer; 2017

Appendix 6

HISTOPATHOLOGY WORKSHEET FOR BREAST BIOPSY/ MASTECTOMY

HSB/PAT/HIS/BOR/01
(Breast Cancer Pin 1/2015)

	KEMENTERIAN KESIHATAN MALAYSIA PERKHIDMATAN PATOLOGI HOSPITAL	HISTOPATHOLOGY WORKSHEET FOR BREAST BIOPSY/MASTECTOMY
Name:		HPE No:
		IC No:

PATHOLOGIST & MO IN-CHARGE:

GROSSING DATE:

NOTES:

SLIDES READY ON:

Received by:	*Blocking:	No of blocks:	*Slides to MO:
Grossed by:	*Sectioning:	No of slides:	*Slides to Pathologist:
Assisted by:	*Staining:	*QC check:	* Please write date and name/initials

SPECIMEN CONTAINER LABELED AS:

1. GROSS DESCRIPTION

- 1.1. Type of specimen: ☐ Mastectomy - ☐ Right ☐ Left
- ☐ Mastectomy with axillary resection - ☐ Right ☐ Left
- ☐ Re-excision (completion mastectomy) - ☐ Right ☐ Left
- ☐ Lumpectomy/excision biopsy (weight: grams)
- ☐ Hookwire Localization biopsy (weight: grams)
- ☐ Wide excision (weight: grams)

- 1.2. Size: Breast tissue:
 Axillary tissue:
 Skin tissue:

- 1.3. Appearance of skin & nipple:

- 1.4. Description of lesion(s)

Focality: ☐ Single ☐ Multiple

Site: ☐ UOQ ☐ UIQ ☐ LOQ ☐ LIQ

☐ Central ☐ _____

Size:

Appearance:

Additional comments:

- 1.5. Distance from tumour to resection margins:

Superior -	Medial -
Inferior -	Lateral -
Deep -	Superficial -

- 1.6. Number of lymph nodes retrieved: (..... mm to mm in diameter)

- 1.7. Sampling (see below)

Grossed by:

2. HISTOLOGY**2.1. Microscopic description:**Histological type: ☐ Invasive carcinoma (NST)☐ Invasive lobular carcinoma☐ Mucinous carcinoma☐ Tubular carcinoma☐ Papillary carcinoma☐ Medullary☐ Other:Tubular score: ☐ 1 ☐ 2 ☐ 3Nuclear score: ☐ 1 ☐ 2 ☐ 3Mitoses: ☐ 1 ☐ 2 ☐ 3

Comment:

2.2. Modified Bloom & Richardson grade: ☐ 1 (score:/9) ☐ 2 (score:/9)☐ 3 (score:/9)**2.3. Tumour size:** - Whole tumour (invasive & DCIS) size: mm

- Invasive tumour size: mm

2.4. Tumour extent: ☐ Localised ☐ Multiple invasive foci**2.5. Vascular/lymphatic invasion:** ☐ Not seen ☐ Present ☐ Possible**2.6. Pathologic response to neoadjuvant therapy (RCPA Australasia 2012):**☐ Not applicable☐ No definite response☐ Partial pathologic response☐ Complete pathologic response

(Refer explanatory notes):

2.7. Associated lesion(s):- DCIS: ☐ No☐ Yes:

- Histologic type: _____

- Nuclear grade: ☐ High ☐ Intermediate ☐ Low- Necrosis ☐ Absent ☐ Present- Microcalcification: ☐ No ☐ Yes- Extensive intraductal component ($\geq 25\%$): ☐ No ☐ Yes- LCIS/Lobular neoplasia: ☐ No ☐ Yes- Other abnormalities: ☐ No ☐ Yes (specify): _____**2.8. Skin and nipple:** - Paget's disease ☐ No ☐ Yes

- Other comments:

2.9. Resection margins (in relation to both invasive and in-situ components):**2.10. Axillary lymph nodes:** Total retrieved -

Number positive -

Extranodal extension: ☐ No ☐ Yes (? no lymph nodes)If single node positive: ☐ Macrometastasis (>2 mm)☐ Micrometastasis (0.2 mm to 2 mm)☐ Isolated tumour cells (<0.2 mm)

Comment on sentinel lymph node (if applicable):

2.11. Immunohistochemical study:- Oestrogen receptor*: ☐ +ve ☐ -ve (..... % of tumour cells)- Progesterone receptor*: ☐ +ve ☐ -ve (..... % of tumour cells)- c-erb B2: ☐ +ve ☐ -ve ☐ Equivocal/weakly positive (2+)*Staining of $\geq 1\%$ of cells of any intensity is considered positive (ASCO/CAP Guidelines).

*For microinvasive tumour, please comment on ER/PR status in DCIS component.

☐ Done on previous biopsy: Please refer to HPE report _____**3. FINAL INTERPRETATION****Source:** Ministry of Health Malaysia. Jemputan Menghadiri Bengkel 'Standardization of Reporting' Anatomi Patologi, 2015. Kota Bharu 28 Julai 201

Appendix 7

MEDICATIONS IN SYSTEMIC THERAPY OF BREAST CANCER AND THEIR COMMON SIDE EFFECTS

MEDICATION	COMMON SIDE EFFECTS	COMMENTS
CHEMOTHERAPY		
Capecitabine	• Cardiotoxicity: uncommon but can be fatal • Gastrointestinal: diarrhoea, nausea, vomiting, mucositis, abdominal pain • Hand-foot syndrome • Liver: Hyperbilirubinemia • General: fatigue/weakness	• Use with caution in patients with history of heart disease • Dose reduction is a must in patients with moderate renal dysfunction • Monitor for hand-foot syndrome
Cyclophosphamide	• Cardiotoxicity: uncommon but can be fatal • Urotoxicity (bladder ulceration, necrosis, fibrosis, contracture): seldom but can be fatal • Haematological: myelosuppression (leukopenia, neutropenia, thrombocytopenia, anaemia) • Gastrointestinal: nausea, vomiting, abdominal discomfort, anorexia, diarrhoea • Dermatological: alopecia, rashes, pigmentation, changes in nails	• Use with caution in patients with history of heart disease • Monitoring of complete blood counts is essential • Used with caution in patients with active urinary tract infection • Advise patient to report urinary symptoms
Docetaxel	• Haematological: neutropenia, anaemia, febrile neutropenia, thrombocytopenia, infections • Dermatological: hypersensitivity, neuropathy, skin reactions, alopecia, nail disorders • Gastrointestinal: dysgeusia (taste disturbance), constipation, anorexia, nausea, vomiting, diarrhoea, mucositis • Respiratory: dyspnoea • General: myalgia, asthenia, pain, fluid retention	• Prophylactic anti-emetic and corticosteroids should be given • Use with caution in patients with history of liver impairment and heart disease • Should not be given to patients with neutrophil counts of <1500 cells/mm³ • Monitor blood count

MEDICATION	COMMON SIDE EFFECTS	COMMENTS
Doxorubicin	- Cardiovascular: cardiotoxicity, i.e. cardiomyopathy, congestive heart failure (seldom, irreversible and can be fatal) - Dermatological: extravasation, skin necrosis, cellulitis, vesication, local phlebitis, thrombophlebitis, reversible alopecia, erythematous streaking along vein proximal to site of injection, phlebosclerosis - Gastrointestinal: nausea, vomiting, mucositis (stomatitis, oesophagitis), diarrhoea, abdominal pain - Haematological: myelosuppression, leucopenia, haemorrhage - General: dehydration, facial flushing (if an injection has been given too rapidly)	- Use with caution in patients with history of heart disease - Cardiac function should be assessed before undergoing treatment and has to be carefully monitored throughout therapy - Contraindicated in patients with marked liver impairment - Initial treatment requires close observation and extensive laboratory monitoring
Epirubicin	- Cardiac toxicity: seldom, irreversible and can be fatal - Haematological: leukopenia, neutropenia, anaemia, thrombocytopenia - Gynaecological: amenorrhea - Gastrointestinal: nausea, vomiting, mucositis, diarrhoea - Dermatological: keratitis, alopecia, local skin toxicity, rash/itch - Ophthalmological: conjunctivitis - General: lethargy, infection	- Use with caution in patients with history of heart disease - Use with caution in patients with hepatic impairment - Cardiac function should be assessed before undergoing treatment and has to be carefully monitored throughout therapy
Eribulin	- Haematological: neutropenia, anaemia - Dermatological: alopecia - Gastrointestinal: nausea, constipation - General: asthenia/fatigue - Peripheral neuropathy	Monitor complete blood counts prior to each dose; increase frequency of monitoring in patients who develop Grade 3 or 4 cytopenias

MEDICATION	COMMON SIDE EFFECTS	COMMENTS
Fluorouracil (5-FU)	- Gastrointestinal: diarrhoea, stomatitis, oesophagitis, heart burn - Haematological: anaemia, leukopenia, thrombocytopenia - Hand-foot syndrome - Cardiovascular: angina pectoris, myocardial infarction, arrhythmia, acute pulmonary oedema - Dermatological: alopecia, dermatitis	- Prophylactic anti-emetic and corticosteroids should be given - Use with caution in patients with history of heart disease - Monitor hand-foot syndrome
Gemcitabine	- Gastrointestinal: nausea, vomiting - Haematological: neutropenia, anaemia, thrombocytopenia - Hepatological: increased liver enzymes - Nephrological: proteinuria, haematuria - Respiratory: dyspnoea - Dermatological: rash - General: fever, oedema	- Monitor for myelosuppression prior to each cycle - Monitor renal function and hepatic function prior to initiation and during treatment
Paclitaxel	- Haematological: anaemia, infections - Dermatological: hypersensitivity reaction, irritation at injection site, alopecia - Gastrointestinal: nausea, vomiting, diarrhoea, mucositis - Neuropathy - General: myalgia, oedema, hypotension	- Patients should be pretreated with corticosteroids, diphenhydramine and H₂ antagonists - Should not be given to patients with neutrophil counts of <1500 cells/mm³ - Monitor blood count
Vinorelbine	- Haematological: neutropenia, anaemia - Liver enzyme elevation - Gastrointestinal: nausea, vomiting, constipation - Dermatological: injection site reaction - Peripheral neuropathy - General: asthenia	- Monitor complete blood counts prior to each dose; do not administer to patients with neutrophil counts <1000 cells/mm³ - Monitor liver function during treatment - Consider routine stool softener during treatment

MEDICATION	COMMON SIDE EFFECTS	COMMENTS
TARGETED THERAPY		
Abemaciclib	• Gastrointestinal: diarrhoea, nausea, vomiting, abdominal pain, decreased appetite • Haematological: neutropenia, leukopenia, thrombocytopenia, anaemia, infections • General: fatigue, headache	• Monitor complete blood counts prior to the start of therapy, every 2 weeks for first 2 months, monthly for the next 2 months and as clinically indicated
Ado-trastuzumab emtansine	• Infusion-related reactions: flushing, chills, pyrexia, dyspnoea, hypotension, wheezing, bronchospasm, tachycardia • Gastrointestinal: nausea, constipation • Haematological: thrombocytopenia • Hepatological: increased liver enzymes • Vascular: haemorrhage, epistaxis • Musculoskeletal pain, peripheral neuropathy • General: headache, fatigue, arthralgia	• Hepatotoxicity and cardiac toxicity: uncommon but can be fatal • Use with caution in patient with history of hepatic impairment and heart disease • Evaluate cardiac function prior to and during treatment
Everolimus	• Gastrointestinal: stomatitis, diarrhoea, abdominal pain, nausea, decreased appetite • Haematological: infections, fever, rash • Respiratory: cough • General: fatigue, asthenia, oedema, headache	• Monitor complete blood count prior to starting and every 6 months for the first year of treatment
Lapatinib	• Gastrointestinal: diarrhoea, nausea, vomiting • Hand-foot syndrome • Dermatological: rash • General: fatigue	• Monitor hand-foot syndrome

MEDICATION	COMMON SIDE EFFECTS	COMMENTS
Olaparib	- Gastrointestinal: nausea, vomiting, abdominal pain, diarrhoea, dysgeusia, dyspepsia, decreased appetite, constipation, stomatitis - Haematological: neutropenia, leukopenia, infections - General: fatigue, asthenia, dizziness	Monitor hematological toxicity at baseline and monthly thereafter
Palbociclib	- Haematological: neutropenia, leukopenia, anaemia, thrombocytopenia, infections - General: fatigue, asthenia, pyrexia - Gastrointestinal: nausea, diarrhoea, vomiting, stomatitis, decreased appetite - Dermatological: alopecia, rash	Monitor complete blood count prior to start therapy and at the beginning of each cycle, on day 15 of the first 2 cycles and as clinically indicated
Pertuzumab	- Left ventricular dysfunction: seldom but can be fatal - Infusion reactions: pyrexia, chills, fatigue, headache, asthenia, hypersensitivity, nausea, vomiting, rash - Gastrointestinal: diarrhoea - Haematological: neutropenia - General: fatigue	Evaluate cardiac function prior to and during treatment
Ribociclib	- Haematological: neutropenia, leukopenia, infections - Gastrointestinal: nausea, vomiting, diarrhoea, constipation - Dermatological: alopecia, rash - Respiratory: cough - General: fatigue, headache	- Use with caution in patients with history of heart problems and hepatic impairment - Perform complete blood count before initiating therapy; monitor it every 2 weeks for the first 2 cycles, at the beginning of each subsequent 4 cycles and as clinically indicated
Trastuzumab	- Cardiotoxicity: can be fatal - Infusion reactions: fever, chills, nausea, vomiting, pain (in some cases at tumour sites), headache, dizziness,	- Evaluate cardiac function prior to and during treatment - Prophylactic antihistamines should be given

MEDICATION	COMMON SIDE EFFECTS	COMMENTS
	dyspnoea, hypotension, rash, asthenia • Gastrointestinal: diarrhoea • Haematological: neutropenia, anaemia, infections • Respiratory: cough • General: fatigue, myalgia	
HORMONAL THERAPY		
Anastrozole	• Dermatological: hot flushes, rash • Musculoskeletal: arthritis, pain, back pain, bone pain, arthralgia, asthenia • Hypertension, headache • Depression, insomnia • Gastrointestinal: nausea, vomiting, pharyngitis • Bone: osteoporosis, fractures • Peripheral oedema, lymphoedema • Respiratory: cough, dyspnoea	• Monitor bone mineral density throughout treatment
Exemestane	• Dermatological: hot flushes, sweating • Gastrointestinal: nausea, increased appetite • General: fatigue	• Monitor bone mineral density
Fulvestrant	• Gastrointestinal: nausea, vomiting, constipation, anorexia • Dermatological: injection site pain, hot flushes • Musculoskeletal: bone pain, back pain, musculoskeletal pain, arthralgia, fatigue • Respiratory: dyspnoea • Increased hepatic enzymes	• Use with caution in patients with hepatic impairment
Letrozole	• Dermatological: hot flushes, flushing • Headache, dizziness • Arthralgia, asthenia • Oedema	• Monitor bone mineral density throughout treatment

MEDICATION	COMMON SIDE EFFECTS	COMMENTS
	- Hypercholesterolaemia - Increased sweating - Bone and musculoskeletal pain	
Tamoxifen	- Gastrointestinal disorders: nausea - General disorder: fatigue/asthenia	Use with caution in patients with history of thromboembolic events

Sources:

1. Xeloda® (capecitabine) [package insert]. South San Francisco, CA: Hoffman-La Roche, Inc. :2015.
2. Endoxan® (cyclophosphamide) [package insert]. Deerfield, IL; Baxter, Inc: 2013.
3. Taxotere® (docetaxel) [package insert]. Bridgewater, NJ: Sanofi-Aventis, Inc: 2019.
4. Adriamycin® (doxorubicin) [package insert]. New York, NY: Pfizer, Inc: 2019.
5. Ellence® (epirubicin) [package insert]. New York, NY: Pfizer, Inc: 2019.
6. Halaven® (eribulin) [package insert]. Woodcliff Lake, NJ: Eisai, Inc: 2017.
7. Micromedex Solutions, Truven Health Analytics Inc. MIMS Gateway Service Portal. (Available at: <http://www.mimsgateway.com/Malaysia/Online.as>)
8. Gemzar® (gemcitabine) [package insert]. Indianapolis, IN: Eli Lilly, Inc: 2019.
9. Paclitaxel Injection, USP (paclitaxel) [package insert]. Lake Forest, IL: Hospira, Inc: 2018
10. Navelbine® (vinorelbine) [package insert]. Boulogne, France: Pierre Fabre, Inc: 2014.
11. Verzenio® (abemaciclib) [package insert]. Indianapolis, IN: Eli Lilly, Inc: 2019.
12. Kadcyia® (ado-trastuzumab emtansine) [package insert]. South San Francisco, CA: Genentech, Inc: 2019.
13. Afinitor® (everolimus) [package insert]. East Hanover, NJ: Novartis, Inc: 2018.
14. Tykerb® (lapatinib) [package insert]. East Hanover, NJ: Novartis, Inc: 2018.
15. Lynparza® (olaparib) [package insert]. Wilmington, DE: AstraZeneca, Inc: 2019.
16. Ibrance® (palbociclib) [package insert]. New York, NY: Pfizer, Inc: 2019.
17. Perjeta® (pertuzumab) [package insert]. South San Francisco, CA: Genentech, Inc: 2018.
18. Kisqali® (ribociclib) [package insert]. East Hanover, NJ: Novartis, Inc: 2019.
19. Herceptin® (trastuzumab) [package insert]. South San Francisco, CA: Genentech, Inc: 2015.
20. Arimidex® (anastrozole) [package insert]. Baudette, MN: ANI Pharmaceuticals, Inc: 2018.
21. Aromasin® (exemestane) [package insert]. New York, NY: Pfizer, Inc: 2019.
22. Faslodex® (fulvestrant) [package insert]. Wilmington, DE: AstraZeneca, Inc: 2019.
23. Femara® (letrozole) [package insert]. East Hanover, NJ: Novartis, Inc: 2014.
24. Nolvadex®-D (ribociclib) [package insert]. Mississauga, Ontario: AstraZeneca Canada, Inc: 2015.

Appendix 8

BRCA1/2 TESTING CRITERIA^{a,b}

Meeting one or more of these criteria warrants further personalised risk assessment, genetic counselling and, often genetic testing and management. Testing of an individual without a cancer diagnosis should only be considered when an appropriate affected family member is unavailable for testing.

- Individual from a family with a known BRCA1/2 pathogenic/likely pathogenic variant, including such variants found on research testing^b
- Personal history of breast cancer^c + one or more of the following:
 - Diagnosed ≤45 years of age
 - Diagnosed 46 - 50 years of age with:
 - An additional breast cancer primary at any age^d
 - ≥1 close blood relative^e with breast cancer at any age
 - ≥1 close blood relative^e with high-grade (Gleason score ≥7) prostate cancer
 - An unknown or limited family history^a
 - Diagnosed ≤60 years of age with:
 - Triple negative breast cancer
 - Diagnosed at any age with:
 - ≥1 close blood relative^e with:
 - breast cancer diagnosed ≤50 years of age; or
 - ovarian carcinoma;^f or
 - male breast cancer; or
 - metastatic prostate cancer;^g or
 - pancreatic cancer
 - ≥2 additional diagnoses^g of breast cancer at any age in patient and/or in close blood relatives
 - Ashkenazi Jewish ancestry^h
- Personal history of ovarian carcinoma^f
- Personal history of male breast cancer
- Personal history of pancreatic cancerⁱ
- Personal history of metastatic prostate cancer^g
- Personal history of high-grade prostate cancer (Gleason score ≥7) at any age with
 - ≥1 close blood relatives^e with ovarian carcinoma, pancreatic cancer or metastatic prostate cancer^g at any age or breast cancer <50 years of age; or
 - ≥2 close blood relatives^e with breast or prostate cancer at any age; or
 - Ashkenazi Jewish ancestry^h

- BRCA1/2 pathogenic/likely pathogenic variant detected by tumour profiling on any tumor type in the absence of germline pathogenic/likely pathogenic variant analysis
- Regardless of family history, some individuals with an BRCA-related cancer may benefit from genetic testing to determine eligibility for targeted treatment^f
- An individual who does not meet the other criteria but with ≥ 1 first- or second-degree blood^e relative^k meeting any of the above criteria. The significant limitations of interpreting test results for an unaffected individual should be discussed.

^a For further details regarding the nuances of genetic counselling and testing, see BR/OV-A.

^b Irrespective of degree of relatedness

^c For the purposes of these guidelines, invasive and ductal carcinoma in situ breast cancers should be included

^d Two breast cancer primaries includes bilateral (contralateral) disease or two or more clearly separate ipsilateral primary tumours either synchronously or asynchronously

^e Close blood relatives include first-, second- and third-degree relatives on same side of family (see BR/OV-B)

^f Includes fallopian tube and primary peritoneal cancers. BRCA-related ovarian cancers are associated with epithelial non-mucinous histology. Lynch syndrome can be associated with both non-mucinous and mucinous epithelial tumours. Be attentive for clinical evidence of Lynch syndrome (see NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal). Specific types of non-epithelial ovarian cancers and tumours can also be associated with other rare syndromes. Examples include an association between sex-cord tumours with annular tubules and Peutz-Jeghers syndrome or Sertoli-Leydig tumors and DICER1-related disorders.

^g Metastatic prostate cancer is biopsy-proven and/or with radiographic evidence and includes distant metastasis and regional bed or nodes. It is not a biochemical recurrence.

^h Testing for Ashkenazi Jewish founder-specific pathogenic/likely pathogenic variant(s), should be performed first. Comprehensive genetic testing may be considered if ancestry also includes non-Ashkenazi Jewish relatives or if other BRCA-related criteria are met. Founder pathogenic/likely pathogenic variants exist in other populations.

ⁱ Approximately 2 - 5% of unselected cases of pancreatic adenocarcinoma will have BRCA1/2 pathogenic/likely pathogenic variant. However, the disease is highly lethal and the option to test the affected relative may not be available in future. Thus, there may be significant benefit to family members in testing these patients near the time of diagnosis. In addition, increasing evidence suggests that identification of BRCA1/2 pathogenic variant may direct use of targeted therapies for patients with pancreatic cancer (see NCCN Guidelines for Pancreatic Adenocarcinoma). (Holter S, Borgida A, Dodd A, et al. J Clin Oncol 2015;33:3124-3129. Shindo K, YU J, Suenaga M, et al. J Clin Oncol 2017;35:3382-3390.

^j E.g., PARP inhibitors for ovarian cancer and metastatic HER2-negative breast cancer; platinum therapy for prostate cancer. See the relevant NCCN treatment guidelines (e.g. NCCN Guidelines for Breast Cancer; NCCN Guidelines for Prostate Cancer) for further details.

^k This may be extended to an affected third-degree relative if related through two male relatives (e.g. paternal grandfather's mother or sister).

Source: National Comprehensive Cancer Network Guidelines Version 3.2019. BRCA-Related Breast and/or Ovarian Cancer Syndrome. NCCN; 2019.

Appendix 9

POST-TREATMENT CANCER SURVIVORSHIP (MANAGEMENT OF TREATMENT COMPLICATIONS)

Breast cancer patients may be at risk of developing treatment-related complications. Thus, healthcare providers must recognise and manage the long-term sequelae of the constellation of therapeutic modalities. There is limited evidence for multidisciplinary rehabilitation among these populations. A systematic review concluded that multidisciplinary rehabilitation was not harmful and may improve functional ability and quality of life in the short-term.¹

- **Breast cancer-related lymphoedema**

Breast cancer-related lymphoedema (BCRL) is a common consequence of breast cancer treatment. Patient should be counselled on ways to prevent or reduce risk of lymphedema. Weight loss for those who are overweight or obese may reduce this risk.^{2,3} Good evidence showed that there was no increased risk of BRCL with exercise; thus patient should not restrict or avoid physical activity.⁴ Patients should be routinely examined for clinical symptoms or swelling suggestive of lymphoedema. They should be referred to therapists knowledgeable about the diagnosis and treatment of it.

- Physiotherapist plays an important role in the rehabilitation care of women with breast cancer as well as the care of the survivors.

- **Late onset cardiotoxicity**

Occurrence of late onset cardiotoxicity associated with chemotherapy is generally low except for anthracycline especially if dose given is above the limit. If occur, it typically presents as reduced left ventricular function failure. In contrast, trastuzumab cardiotoxicity usually occur during treatment and does not have risk of delayed cardiotoxicity.⁵

Identification of high-risk patients including patients with pre-existing heart problems, cardiovascular (CV) risk factors and treatments like trastuzumab and anthracyclines,⁶ education of patients on healthy lifestyle modifications, aggressive management of underlying CV risk factors, consideration of cardioprotective strategies and, routine surveillance of left ventricular function before and after therapies are recommended to reduce breast cancer treatment-associated cardiotoxicities.⁷ Aerobic exercise is considered a promising non-pharmacological strategy to prevent and/or treat chemotherapy-induced cardiotoxicity.⁵

• Cognitive impairment

Cognitive impairment is one of the frequent complications reported by patients with breast cancer. All patients need to be screened for cognitive impairment and to identify reversible contributing factors e.g. organic brain disease, mood disorder, endocrine problem, dehydration, infections or medication-related and optimally treat when possible.

• Fatigue

Studies of long-term cancer survivors suggested that approximately one-quarter to one-third experienced persistent fatigue for up to 10 years after cancer diagnosis.⁸ Fatigue has a negative impact on work, social relationship, mood and daily activities. It causes impairment in overall quality of life during and after treatment. All patients should be assessed for fatigue and treated for any causative factors including anaemia, cardiac dysfunction, depression, sleep disturbance, etc. Patients without otherwise identifiable cause are encouraged to engage in regular physical activity and cognitive behavioural therapy as appropriate.

• Pain

Approximately 20 - 50% of patients complain of pain during the survivorship period. Types of pain may be nociceptive, neuropathic or mixed pattern.^{9,10} Pain will lead to physical, emotional, spiritual and functional discomfort. Pain can be assessed with a comprehensive history taking and simple pain scales e.g. visual analog scale. Aetiology of the pain needs to be identified before initiation of treatment. Refer to CPG on Management of Cancer Pain.¹¹

• Bone health

Survivors of breast cancer are at high risk of significant bone loss. Assessment of bone mineral density in high risk patients as detailed below are:¹²

- post-menopausal survivors should have a baseline DEXA scan
- a repeat DEXA scans every two years should be done for:
 - women taking AIs
 - pre-menopausal women on GnRHa
 - women who have chemotherapy-induced premature menopause

Non-pharmacologic interventions including lifestyle changes, vitamin D and calcium supplements are extremely important.¹³

Source:

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LIST OF ABBREVIATIONS

3D	three-dimensional
¹⁸ F-DG	(¹⁸ F)-fluorodeoxyglucose
ACR	American College of Radiology
AGREE	Appraisal of Guidelines for Research and Evaluation
Als	aromatase inhibitors
ALND	axillary lymph nodes dissection
ASR	Age-Standardised Incidence Rate
AUC	area under the curve
BCRL	Breast cancer-related lymphoedema
BCS	breast conserving surgery
BCSS	breast cancer-specific survival
BI-RADS®	Breast Imaging Reporting and Data System
BOADICEA	Breast and Ovarian Analysis of Disease Incidence and Carrier Estimation Algorithm
BR	breast reconstruction
BRCA	BRest CAncer gene
CBE	clinical breast examination
CDK	cyclin-dependent kinase
CI	confidence interval
cm	centimetre
cm ³	centimetre square
COH	controlled ovarian hyperstimulation
CPG(s)	clinical practice guidelines
CT	computed tomography
CV	cardiovascular
DBT	digital breast tomosynthesis
DCIS	ductal carcinoma in situ
DEXA	dual-energy X-ray absorptiometry
DFS	disease free survival
DMFS	distant metastasis-free survival
DOR	diagnostic odds ratio
ER(-/+)	estrogen receptor (negative/positive)
FFDM	full-field digital mammography
FNAC	fine needle aspiration cytology
FP	fertility preservation
g	gramme
GnRHa	Gonadotropin-releasing hormone agonists
IBTR	ipsilateral breast tumour recurrences
IHC	immunohistochemistry
ISH	in-situ hybridisation
IV	intravenous
HER2(-/+)	Human Epidermal Growth Factor Receptor 2 (negative/positive)
HR	hazard ratio
LA	luminal A
LABC	locally advanced breast cancer
LB	luminal B
LCIS	lobular carcinoma-in-situ
LR	local recurrence
LRR	locoregional recurrence
MBC	metastatic breast cancer
MCG	Mainstreaming Cancer Genetics

MDT	multidisciplinary team
MET	metabolic equivalent of task
MIBT	minimally invasive biopsy technique
mg	milligramme
MoH	Ministry of Health
MRI	magnetic resonance imaging
mTOR	mammalian target of rapamycin
NACT	neoadjuvant chemotherapy
NET	neoadjuvant endocrine therapy
OAS	ovarian ablation or suppression
OC	oral contraceptives
ONJ	osteonecrosis of the jaw
OR	odds ratio
OS	overall survival
PBI	partial breast irradiation
PET	positron emission tomography
PFS	progression free survival
PMRT	post-mastectomy radiation therapy
PNP	patient navigation programme
PR(-/+)	progesterone receptor (negative/positive)
RCT(s)	randomised controlled trial(s)
RGCT	rapid genetic counselling and testing
RR	relative risk
RS	relative survival
RRM	risk-reducing mastectomy
RRSO	risk-reducing bilateral salpingo-oophorectomy
RT	radiotherapy
SC	subcutaneous
SERD	selective estrogen receptor degrader
SLND	sentinel lymph nodes dissection
SLNs	sentinel lymph nodes
SREs	skeletal-related events
SRS	stereotactic radiosurgery
TE	tissue expander
TNM	Tumour Node Metastasis
US	ultrasound
vs	versus
WBI	whole breast irradiation
WBRT	whole brain radiotherapy
WHO	World Health Organization

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